

Arnav Kumar

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EDUCATION

The Ohio State University

B.S. in Computer Science

Columbus, OH

Aug. 2024 – May 2028

EXPERIENCE

ISO Counsel Intern

CyberTryzub (US-based Cybersecurity Firm)

Jun. 2025 – Jul. 2025

Remote

- Participated in cybersecurity audit and advisory calls with **startups to assess organizational risk exposure** and compliance readiness.
- Supported **ISO 27001 documentation** review and identified potential security and compliance gaps from client evidence.

Research Assistant

Ohio State University (Prof. Brijesh Soni)

Mar. 2026 – Present

Columbus, OH

- Conducting research on **structured pruning and quantization for efficient deployment of LLMs** using small language models including Qwen2.5, Torch-Pruning, and llama.cpp.
- Built a **GPU-accelerated local inference and QA benchmarking pipeline** across SQuAD, BoolQ, and Natural Questions using llama-server.
- Improved SQuAD evaluation performance to approximately **44% EM / 52% F1** through prompt engineering, answer extraction, and evaluation-pipeline optimization.

Research Intern

Kyoto University of Advanced Sciences (Dr. Zilu Liang)

May 2026 – Present

Kyoto, Japan

- Conducting machine learning research on **personalized post-meal glucose response prediction** using wearable, biomarker, and meal data under a **LOSO evaluation framework** across 41 participants.
- Built and evaluated **XGBoost, CatBoost, Random Forest, and Gradient Boosting** pipelines for glucose-response metrics including AUC and iAUC.
- Implemented leakage-aware nested cross-validation using **GroupKFold** and **RandomizedSearchCV** for subject-independent hyperparameter tuning.
- Achieved approximately **$R^2 \sim 0.60$** for AUC prediction while identifying substantially lower generalization performance for iAUC prediction.

PROJECTS

Energy Forecasting with Anomaly Detection | Python, pandas, scikit-learn, XGBoost

Aug. 2025

- Built a time-series forecasting pipeline on **2M+ readings**, engineering **25 temporal features**.
- Applied anomaly detection (**Z-score, IQR, Modified Z-score**) to clean data and improve model reliability.
- Evaluated regression models (Linear, Decision Tree, XGBoost); best model achieved **MAE 0.0123, RMSE 0.0200**.

AI-Powered Image Editor | Python, DeepFace, ResNet, Gradio

Jul. 2025

- Built an interactive image editing tool integrating **DeepFace facial analysis and ResNet backbones**.
- Designed a Gradio interface enabling interactive editing and chatbot-style user interaction.

Mg-Based Nanorobots in Oncology (Honors Project) | ENGR 1282.02H

Apr. 2025

- Led a 3-person team project and authored a **research paper on magnesium-based nanorobots** for targeted cancer drug delivery.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, HTML/CSS

ML / Data: TensorFlow/Keras, NumPy, pandas, scikit-learn, Matplotlib, Seaborn

Tools: Git/GitHub, ONNX, OpenCV, Gradio, MATLAB, SolidWorks

Concepts: Deep Learning, Time-Series Modeling, Model Optimization, Data Engineering

Coursework: Coursework: Machine Learning Specialization (Andrew Ng, Stanford/Coursera)